DATA7201 Data Analytics at Scale

Project Report

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1. Abstract

This project analyses strategic political advertising on Facebook in Australia from March 2020 to February 2024, using data from the Facebook Ad Library. The original dataset comprised over 39 million ad snapshots, which were extensively cleaned and deduplicated to produce a final set of 482,000 unique political ad messages. Using PySpark on an HDFS-based cluster, scalable preprocessing, keyword extraction, and spend analysis were performed. Key findings include heavy ad spend by a few funders, notably Greenpeace Australia Pacific and Australian Labor party with peaks during election periods and the 2023 Voice referendum. COVID-related ads accounted for over $2 million in spend, showing consistent messaging across pandemic phases. This study demonstrates how big data tools can uncover patterns in political messaging, funder influence, and temporal trends, offering insights into how emotional and event-driven communication strategies are deployed in digital political campaigns. (Katal, Wazid, & Goudar, 2015)

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3. Introduction

Facebook has become a central platform for digital political campaigning, offering precise targeting and widespread reach. To promote transparency, Facebook provides access to its Ad Library API, which archives metadata on paid political ads, including funding sources and demographic reach. This project analyses Facebook political ad data targeting Australian users between March 2020 and February 2024, a period that includes the 2022 federal election and the 2023 Voice referendum. Using PySpark on a distributed HDFS cluster, the analysis explores strategic ad placement through keyword usage, funding concentration, and temporal patterns. A special focus was given to COVID-related messaging to assess its political utility across different phases of the pandemic. This report presents insights into how data-driven advertising strategies align with public events and policy debates, illustrating the power of big data analytics in uncovering patterns in political communication.

4. Dataset Analytics

4.1 Dataset Description and Preprocessing

The original dataset contained over 39.5 million JSON-based Facebook Ad snapshots (March 2020–February 2024), stored at "/data/ProjectDatasetFacebookAU/fbadAU/". These included variable ad structures — *ad_creative_body* (string) and *ad_creative_bodies* (array) — depending on how ads were created. Snapshots missing key fields (*ad_creative_bodies*, *funding_entity*, or *page_name*) were removed, with a failsafe to retain those with only *ad_creative_body*. Arrays were exploded into individual ads, and *spend.upper_bound* and *impressions.upper_bound* were cast to integers. Duplicates were dropped using ad IDs. The final dataset comprised 482,228 unique ad messages with consistent fields (date, text, funder, spend, impressions, page source), enabling robust keyword and spend analysis. (Katal, Wazid, & Goudar, 2015)

4.2 Keyword Frequency Analysis

To determine the dominant messaging, a combination of regular expression tokenization and normalization to lower case text, where common stop words were removed. Note that the distribution showed tremendous use of words with emotional and persuasion meanings, and phrases such as "vote", "support", "you" and "community" showing a rhetorical strategy associated with connection and urgency. Overall, the results verify, but primarily that political

messaging was based upon the use of emotionally charged language. (Bird.S, Klein.E, & Loper.E., 2009)

4.3 Spending by Keyword

After preprocessing and tokenizing ad content, keywords were extracted and filtered using a comprehensive custom stop-word list to highlight meaningful campaign terms. High-spend keywords included politically significant terms tied to the 2023 Voice referendum, such as "yes," "voice," "indigenous," "vote," "pledge," "support," and "recognition," indicating heavy financial focus on this campaign. Additionally, grouped spending around keywords like "aboriginal," "islander," and "referendum" revealed organised, partisan messaging. The analysis demonstrates a deliberate language strategy, with financial backing reinforcing targeted political narratives.

4.4 Top Funders and Pages

Spending across the 482,000 cleaned ads was heavily concentrated among a few funders, led by Greenpeace Australia Pacific, followed by the Australian Labor Party, United Australia Party, Victoria Labor, Amnesty International Australia, and Yes23. Page-level analysis showed that the highest-spending Facebook pages aligned closely with these funders. While major political and advocacy groups dominated ad visibility, several civic-focused pages also received notable funding. This highlights both the strategic concentration and diversity in communication avenues for political messaging on Facebook.

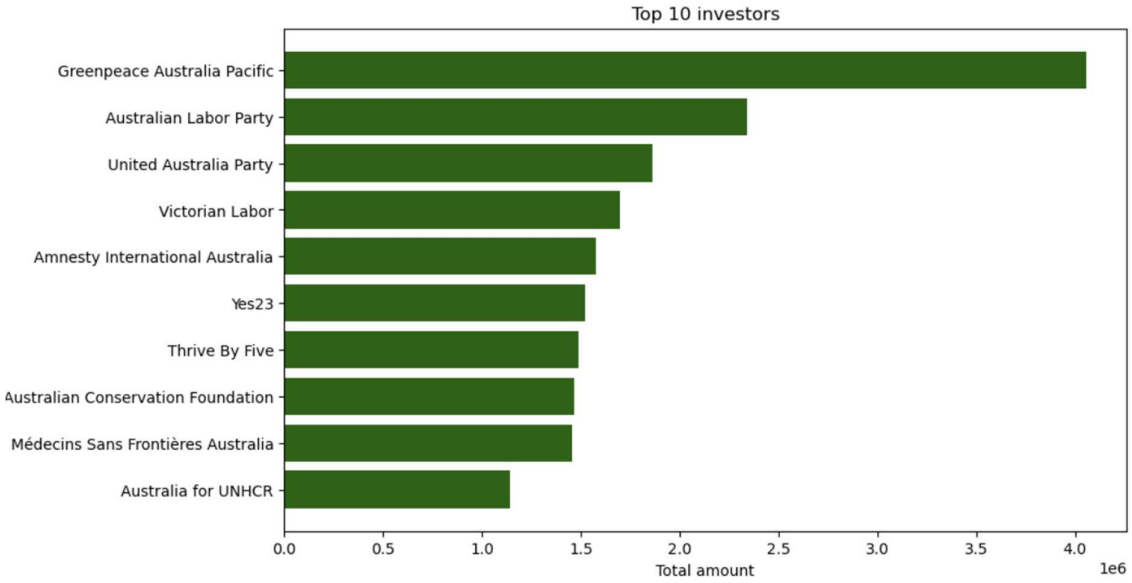

Figure 1. Top 10 funding entities ranked by total estimated Facebook ad spend.

The bar chart (Figure 1) Top 10 funding entities by total Facebook ad spend. Political and advocacy groups like Greenpeace, Labor, and United Australia Party, were key players for total ad investments.

4.5 Time-based trends

The analysis of advertising showed a cyclical period of activity, encapsulated with spikes indicating election times and significant public milestones. Daily ad volume and total spend per day indicate significant spikes around late 2022 and further spikes mid to late 2023, with associated national campaign and referendum-inspired efforts. There were also spikes in 2020 and 2021 that coincided with Government communication periods during pandemic times. This finding supports how political and advocacy groups can tactically plan their campaigns to coincide with heightened public interest. The overlapping of expenditure spikes with civic moments around budgeting effort shows the campaign deployment of advertising messages was planned accordingly.

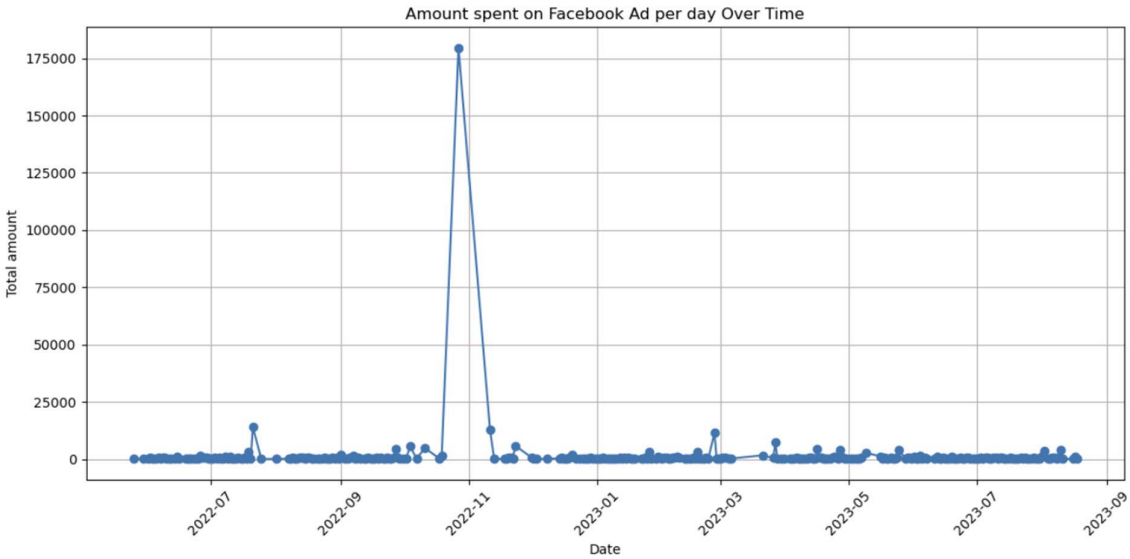

Figure 2. Daily Facebook political ad spends from March 2020 to February 2024.

Figure 2 shows Daily Facebook ad spend from 2020 to 2024, with visible peaks near elections and major national events.

4.6. COVID Deep Dive

An intentional analysis explored how the pandemic shaped political and advocacy Facebook ad messaging. Over 14,000 ads containing "covid" were identified, collectively spending over $2M — confirming that pandemic messaging was deliberate and sustained across campaign phases from early 2020 through late 2022. While 2023 lacks spend data, ad volumes peaked during key moments like the first COVID wave and vaccination drives, suggesting real-time responses to public sentiment. Earlier ads showed concentrated messaging on pandemic values, revealing attempts to shape public reactions during crises.

Funders such as Amnesty International Australia, United Australia Party, and UNICEF dominated this category, with messaging ranging from civic cooperation and health safety to political critique. Keyword analysis echoed themes like "community," "support," "freedom," and "economic recovery." These findings confirm COVID-19 was central to campaign strategy — not background noise but a lever for emotional and values-based appeals. The pandemic became a strategic tool for building trust, urgency, or opposition across mainstream and fringe campaigns.

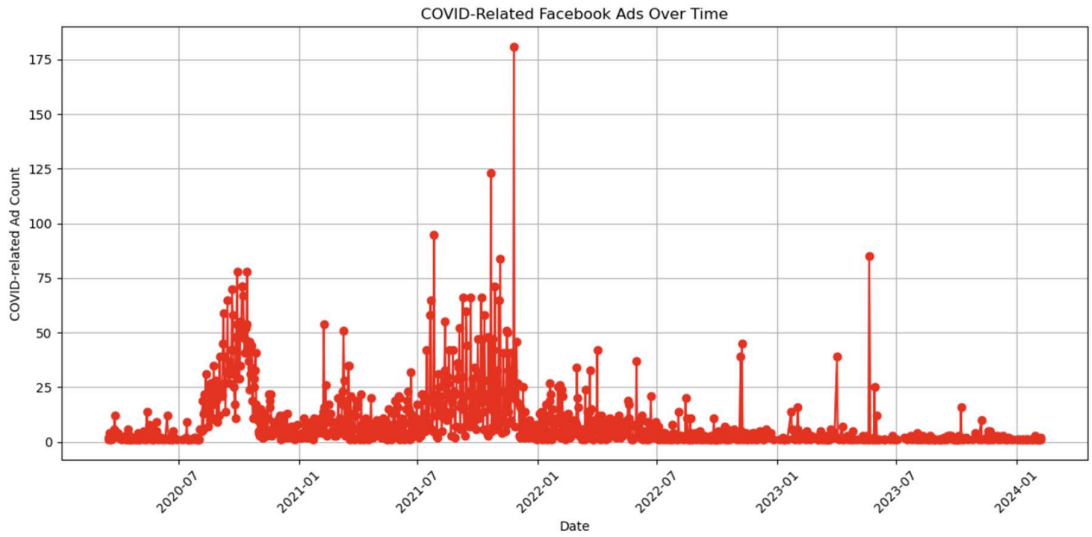

Figure 3. Daily Facebook political ad spends from March 2020 to February 2024.

Figure 3 shows the Daily volume of Facebook ads tagged with COVID from 2020 to 2024. Daily ad spikes, frequently aligned with major pandemic phases, demonstrate the degree of responsiveness campaigns had to prevailing national sentiment.

4.7 Cost Efficiency - CPM Analysis

To assess the cost effectiveness of each ad, a CPM (cost per 1,000 impressions) metric was generated by combining the upper-bound spends to impressions. Most ads were very cost effective in their delivery with more than 90% of ads having CPM's lower than $50. A notable cluster around $99 may indicate platform rounding or a default limit on high-cost placements. The lowest CPMs were observed in humanitarian campaigns, most notably UNICEF campaigns, that had CPMs frequently under $1, indicative of wide reach and low targeting costs. Conversely, ads about referendum campaigns and narrowly targeted political ads tended to have CPMs upwards of $50, and in some cases over $199 CPMs were driven by low impression counts and narrow targeting. Despite these trends, with relatively large-scale advocacy campaigns focused on broad awareness and intensity, political focused campaigns demonstrated an intensity without reach. CPMs proved to be reliable indicators in distinguishing high-engagement campaigns from more narrowly targeted efforts.

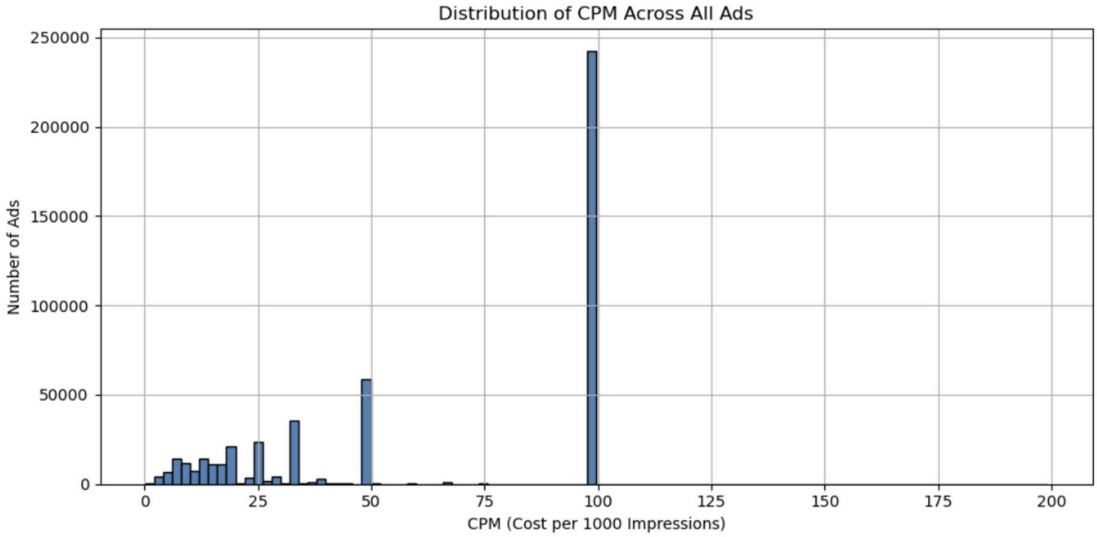

Figure 4: Distribution of CPM (Cost per 1,000 Impressions) Across Facebook Ads

Figure 4 shows the Distribution of CPM values across all ads. Most ads were under $50, with a sharp spike near $99, indicating there were cost ceiling effects.

5. Discussion and Insights

The analysis of 480,000+ Facebook political ads revealed patterns aligning with elections, public health phases, and the 2023 Voice referendum. Financial resources were concentrated among a few funders like Greenpeace, Labor, and the United Australia Party. COVID-19 was a recurring theme used for varied messaging from civic engagement to political criticism.

The findings revealed how political communication on Facebook mirrors national events, with campaigners using emotional messaging to boost visibility. While humanitarian ads (e.g., UNICEF) were highly cost-efficient, political and referendum ads tended to prioritize targeted reach over broad exposure, resulting in higher CPMs.

The results demonstrate that ad strategy involves more than messaging — it includes timing, emotional framing, and budget targeting. This highlights Facebook's continuing role as a strategic platform in digital political discourse.

6. Tools And Technology

All data processing was performed using PySpark on HDFS machines, enabling scalable operations like filtering, flattening arrays, text tokenization, and spend aggregation. Visual summaries such as keyword distributions, funder rankings, and trend plots were generated directly on the cluster using Matplotlib and Pandas. This integration allowed low-level data processing to directly inform high-level insights, supporting distributed big data analysis. (Zaharia, Chowdhury, Das, Dave, & Justin Ma, 2018)

7. Conclusion

This project explored more than 480,000 cleaned political ad messages out of a total sample that exceeded 39 million Facebook ad snapshots. Distributed PySpark processing revealed how ad campaigns were executed and funded and how they were strategically timed.

COVID-19 figured prominently within the various phases of campaign activity. More than 2 million AUD was spent on COVID-related ads, which included messages designed to mobilize people around safety and civic engagement or to criticize particular political actors based on how they responded to the pandemic. Cost-effective analysis revealed that humanitarian campaigns reached more per dollar spent than both referendum and political ads that tended to spend more per impression.

Overall, the study's findings demonstrated that a small number of funders dominate the digital political landscape, take advantage of emotional themes, deliver ads based on event-driven timing, and being conscious of budgets in the delivery of ads. Facebook remains a powerful technological platform for generating and contouring political discourse in Australia with an emphasis on adaptability.

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Appendix

```
from pyspark.sql import SparkSession
from pyspark.sql.functions import (
    col, when, array, explode, split, lower, regexp_replace, count, sum
)
import matplotlib.pyplot as plt
import pandas as pd

#All the ads are loaded in datafile
spark = SparkSession.builder.getOrCreate()
datafile = spark.read.option("multiLine",
"false").json("hdfs:///data/ProjectDatasetFacebookAU/fbadsAU/*")
datafile.cache()

#Filtering out Null values from the datafile
ads = datafile.filter(
    (col("funding_entity").isNotNull()) &
    (col("page_name").isNotNull()) &
    ((col("ad_creative_bodies").isNotNull()) | (col("ad_creative_body").isNotNull()))
)

ads = ads.dropDuplicates(["id"])

ads = ads.withColumn(
    "text_array",
    when(col("ad_creative_bodies").isNotNull(), col("ad_creative_bodies"))
    .otherwise(array(col("ad_creative_body")))
)
```

```
ads_flat = ads.select(
    "ad_creation_time",
    explode("text_array").alias("ad_text"),
    "page_name",
    "funding_entity",
    col("spend.upper_bound").cast("int").alias("spend_upper"),
    col("impressions.upper_bound").cast("int").alias("impressions_upper")
)

ads_clean = ads_flat.filter((col("ad_text") != "") & col("ad_text").isNotNull())

#Slpitting all the ad texts into words
ads_split = ads_clean.withColumn("word", explode(split(lower(col("ad_text")), "\\W+")))

#Filtering out empty strings and basic stop-words
word_filter = ads_split.filter(
    (col("word") != "") &
    (~col("word").isin("the", "and", "to", "a", "of", "is", "in", "on", "for"))
)

# Counting each word's frequency among the filtered words.
word_count = word_filter.groupBy("word").count().orderBy("count", ascending=False)

ads_split_clean = ads_split.drop("spend_upper")

spend_data = ads_clean.select("ad_text", col("spend_upper").alias("spend_amt"))

#Joining the words of ads dataframe and the spend coloumn that has the amount spent on the ad data
join_spend = ads_split_clean.join(spend_data, on="ad_text", how="left")
```

```
#filtering the stop-words
stopwords = [
    "a", "about", "above", "after", "again", "against", "all", "am", "an", "and",
    "any", "are", "aren't", "as", "at", "be", "because", "been", "before", "being",
    "below", "between", "both", "but", "by", "can", "cannot", "could", "couldn't",
    "did", "didn't", "do", "does", "doesn't", "doing", "don't", "down", "during",
    "each", "few", "for", "from", "further", "had", "hadn't", "has", "hasn't",
    "have", "haven't", "having", "he", "he'd", "he'll", "he's", "her", "here",
    "here's", "hers", "herself", "him", "himself", "his", "how", "how's", "i",
    "i'd", "i'll", "i'm", "i've", "if", "in", "into", "is", "isn't", "it", "it's",
    "its", "itself", "let's", "me", "more", "most", "mustn't", "my", "myself",
    "no", "nor", "not", "of", "off", "on", "once", "only", "or", "other", "ought",
    "our", "ours", "ourselves", "out", "over", "own", "same", "shan't", "she",
    "she'd", "she'll", "she's", "should", "shouldn't", "so", "some", "such",
    "than", "that", "that's", "the", "their", "theirs", "them", "themselves",
    "then", "there", "there's", "these", "they", "they'd", "they'll", "they're",
    "they've", "this", "those", "through", "to", "too", "under", "until", "up",
    "very", "was", "wasn't", "we", "we'd", "we'll", "we're", "we've", "were",
    "weren't", "what", "what's", "when", "when's", "where", "where's", "which",
    "while", "who", "who's", "whom", "why", "why's", "with", "won't", "would",
    "wouldn't", "you", "you'd", "you'll", "you're", "you've", "your", "yours",
    "yourself", "yourselves", "s", "d", "t", "m", "re", "ll", "ve"
]

#cleaning the merged words and spend data
join_spend_clean = join_spend.filter(
    (col("word") != "") &
    (~col("word").isin(stopwords)) &
    (col("spend_amt").isNotNull())
)
```

```
# Grouping by total spend based words
word_spend = join_spend_clean.groupBy("word") \
    .sum("spend_amt") \
    .withColumnRenamed("sum(spend_amt)", "total_spend") \
    .orderBy("total_spend", ascending=False)

#Grouping by the top investors that have funded the ADs
investors = ads_clean.groupBy("funding_entity") \
    .sum("spend_upper") \
    .withColumnRenamed("sum(spend_upper)", "total_spend") \
    .orderBy("total_spend", ascending=False)

# Grouping by the top pages that are funded and the total fundings
fb_pages = ads_clean.groupBy("page_name") \
    .sum("spend_upper") \
    .withColumnRenamed("sum(spend_upper)", "total_spend") \
    .orderBy("total_spend", ascending=False)

ad_vol = ads_clean.groupBy("ad_creation_time") \
    .agg(count("*").alias("ad_count")) \
    .orderBy("ad_creation_time")
spend_perday = ads_clean.groupBy("ad_creation_time") \
    .agg(sum("spend_upper").alias("daily_spend")) \
    .orderBy("ad_creation_time")
ads_covid = ads_clean.filter(lower(col("ad_text")).contains("covid"))
ads_covid.select("ad_creation_time", "funding_entity", "page_name", "ad_text",
"spend_upper").show(5, truncate=False)
spend_covid = ads_covid.agg(sum("spend_upper").alias("covid_spend"))

investors_covid = ads_covid.groupBy("funding_entity") \
    .sum("spend_upper") \
```

```
    .withColumnRenamed("sum(spend_upper)", "covid_spend") \
    .orderBy("covid_spend", ascending=False)

ad_vol_covid = ads_covid.groupBy("ad_creation_time") \
    .count() \
    .orderBy("ad_creation_time")

words_covid = ads_covid.withColumn("word", explode(split(lower(col("ad_text")), "\\W+"))) \
    .filter((col("word") != "") &
            (~col("word").isin("the", "and", "to", "a", "of", "is", "in", "on", "for", "you", "your", "this")))

word_count_covid = words_covid.groupBy("word").count().orderBy("count", ascending=False)

#Data frame with spend and impressions
reach_df = ads_clean.select(
    "ad_text",
    "funding_entity",
    "page_name",
    col("spend_upper").cast("double").alias("spend"),
    col("impressions_upper").cast("double").alias("impressions")
).filter(
    col("spend").isNotNull() & col("impressions").isNotNull() & (col("impressions") > 0)
)

#cpm
reach_df = reach_df.withColumn("cpm", (col("spend") / col("impressions")) * 1000)

#displaying top 10 ads with highest cpm
high_cpm_ads = reach_df.filter(col("cpm") > 100).orderBy(col("cpm").desc())
high_cpm_ads.select("funding_entity", "page_name", "spend", "impressions", "cpm").show(10,
truncate=False)
```

```
#displaying top 10 ads with lowest cpm

low_cpm_ads = reach_df.filter(col("cpm") < 10).orderBy(col("cpm"))

low_cpm_ads.select("funding_entity", "page_name", "spend", "impressions", "cpm").show(10,
truncate=False)


#Histogram of CPM values

reach_sample = reach_df.select("cpm").dropna().toPandas()

plt.figure(figsize=(10, 5))

plt.hist(reach_sample["cpm"], bins=100, color='red', edgecolor='black')

plt.xlabel("CPM (Cost per 1000 Impressions)")

plt.ylabel("Number of Ads")

plt.title("Distribution of CPM Across All Ads")

plt.tight_layout()

plt.show()
```